FSDD: STRUCTURAL DAMAGE DETECTION IN POST DISASTER AREA USING UAV IMAGERY

A PROJECT REPORT

submitted by

R.SASANA

P.RAKSHITHA

F.JOEVITA FAUSTINA DOSS

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

DEPARTMENT OF COMPUTER TECHNOLOGY

ANNA UNIVERSITY, MIT CAMPUS

MAY 2023

DEPARTMENT OF COMPUTER TECHNOLOGY

ANNA UNIVERSITY, MIT CAMPUS

CHROMEPET, CHENNAI – 600 044

BONAFIDE CERTIFICATE

Certified that the bonafide thesis report of the project work done by Ms. R.SASANA, Ms. P.RAKSHITHA, Ms. F.JOEVITA FAUSTINA DOSS of Register Number 2020503542, 2020503024, 2020503013 in the Creative and Innovative Project Laboratory subject code CS6611 during the period January 2023 to May 2023

SUPERVISOR	**COURSE IN-CHARGE**
Dr. R. GUNASEKARAN	Dr. S. NEELAVATHY PARI
Professor	Assistant Professor (SI.Grade)
Department of Computer Technology	Department of Computer Technology
Anna University, MIT Campus	Anna University, MIT Campus
Chromepet, Chennai 600 044	Chromepet, Chennai 600 044

ABSTRACT

Natural or man-made disasters can have devastating consequences, causing extensive damage to buildings and infrastructure and endangering the safety of people and property. Identifying the extent and severity of the damage is crucial to assess the situation and prioritize response and recovery efforts. To address these challenges, a novel approach to structural damage detection in post-disaster environments using Unmanned Aerial Vehicles (UAVs) and Deep Learning (DL) is proposed. The proposed Feature-based Structural Damage Detection (FSDD) technique is based on DL, which enables it to learn and detect patterns in the UAV imagery with high accuracy and reliability. The use of UAVs for post-disaster damage assessment is becoming increasingly popular due to their flexibility, mobility, and ability to capture high-resolution images of the affected areas. The FSDD technique leverages this technology to automate the damage detection process and minimize the risks associated with manual inspections. The evaluation of the Instance Segmentation in Building Damage Assessment (ISBDA) dataset shows that the FSDD technique can detect different types of structural damage, including cracks, fractures, and displacements, with high precision and recall. The technique can also quantify the severity of the damage and provide a priority ranking for the buildings that require immediate attention. The proposed approach has the potential to improve the efficiency and effectiveness of post-disaster response and recovery operations by providing accurate and timely information about the state of the infrastructure and the safety of the people. It can also reduce the costs associated with manual inspections and enable faster decision-making in critical situations.

ACKNOWLEDGEMENT

We take this opportunity to thank the respected Dean, Anna University, MIT Campus, **Dr. J. PRAKASH**, and our respected Head of the Department, **Dr. P. JAYASHREE**, Professor and Head of Department, Department of Computer Technology, Anna University, MIT Campus, for providing the lab facilities in pursuit of this project.

Undertaking this project has helped us learn a lot, and we would like to express our sincere gratitude to our supervisor **Dr. R. GUNASEKARAN**, Professor, Department of Computer Technology, Anna University, MIT Campus, whose kind guidance and proper directions helped shape this project perfectly. Constant encouragement to implement the project to the fullest of our potential led us through the difficult phase of the project and assisted in the successful completion of the proposed project work.

We acknowledge the efforts of the panel members **Dr. P. JAYASHREE, Dr. RAJA. KATHIROLI, Dr. V P JAYACHITRA,** in reviewing our work, providing constant valuable comments, and encouraging us to view the different aspects of the project in the successful implementation of the project.

Finally, we thank all the teaching and non-teaching members, Department of Computer Technology, Anna University, MIT Campus, colleagues, NGN Lab scholars, seniors, juniors, and friends whose appreciable help, either directly or indirectly helped in the smooth completion of the project in a limited time frame and made the demanding time joyful but always efficient.

R.SASANA (2020503542)

P.RAKSHITHA (2020503024)

F.JOEVITA FAUSTINA DOSS (2020503013)

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	**iii**
	LIST OF TABLES	**ix**
	LIST OF FIGURES	**x**
	LIST OF ABBREVIATIONS AND NOMENCLATURE	**xi**
1.	**INTRODUCTION**	**1**
	1.1 STRUCTURAL DAMAGE DETECTION	1
	1.1.1 DEEP LEARNING	2
	1.1.2 MACHINE LEARNING	2
	1.1.3 OBJECT DETECTION	3
	1.1.4 ACTIVE LEARNING	3
	1.1.5 ENSEMBLE LEARNING	4
	1.1.6 UNMANNED AERIAL VEHICLES	4
	1.2 OVERVIEW OF PROPOSED WORK	5
	1.3 PROBLEM STATEMENT	6
	1.4 OBJECTIVES	6

2. LITERATURE SURVEY 8

3. PROPOSED SYSTEM 13

3.1 OVERVIEW OF PROPOSED SYSTEM 13

3.1.1 FEATURE EXTRACTION 14

3.1.2 PREPROCESSING & ANNOTATION 15

3.2 ARCHITECTURE OF THE FSDD SYSTEM 16

3.2.1 YOLOv8 CNN-MODEL 16

3.2.2 DETECTRON2 17

3.2.3 YOLOv5 18

3.3 ALGORITHM FOR THE PROPOSED FSDD SYSTEM 20

3.4 OPTIMIZATION AND ACTIVE LEARNING 21

3.5 ENSEMBLE THE MODELS 21

4. IMPLEMENTATION 22

4.1 TOOLS AND LANGUAGES USED 22

4.1.1 PYTHON 22

4.1.2 OPENCV 22

4.1.3 GOOGLE COLLAB 23

4.1.4 CUDA 23

4.1.5 MATPLOTLIB 23

4.1.6 ROBOFLOW 23

4.2 EXPERIMENTAL SETUP 24

4.3 EVALUATION METRICS 24

4.3.1 TRAIN & TEST SET 24

4.3.2 mAP 25

4.3.3 PRECISION AND RECALL 25

4.3.4 F1-SCORE 25

4.3.5 AVERAGE LOSS 25

4.4 COMPARISON OF THE PERFORMANCE OF VARIOUS DL MODELS & FSDD SYSTEM 26

5.	**RESULTS AND ANALYSIS**	**29**
6.	**FUTURE WORKS**	**30**
7.	**CONCLUSION**	**31**
	REFERENCES	**32**

LIST OF TABLES

TABLE NO	TITLE	PAGE NO.
4.1	Performance comparison of various CNN & FSDD system	27

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO
1.1	Overall description of the proposed FSDD system	5
3.1	Building Damage Detection using FSDD system	14
3.2	High-level architecture of the FSDD system	16
4.1	Evaluation Metric	26
4.2	Performance Metrics Comparison	28
4.3	Testing of the YOLOv8 Model	29
4.4	FSDD Model predicted result	29

LIST OF ABBREVIATIONS AND NOMENCLATURE

AI	Artificial Intelligence
mAP	Mean Average Precision
CNN	Convolutional Neural Network
DL	Deep Learning
ML	Machine Learning
R-CNN	Region-based Convolutional Neural Network
UAV	Unmanned Aerial Vehicle
YOLO	You Only Look Once

LIST OF SYMBOLS

λ	Hyperparameter that controls the balance between the two loss components
σ	Sigmoid Function

CHAPTER 1

INTRODUCTION

1.1 STRUCTURAL DAMAGE DETECTION

Structural damage detection is a crucial area of study, especially in the aftermath of natural disasters such as earthquakes, hurricanes, and floods. Such calamities can have a devastating impact on buildings, bridges, and other infrastructure, endangering people's lives. Structural damage detection plays a vital role in identifying buildings that are in danger of collapsing or being damaged, enabling people to evacuate in time. The severity and extent of damage caused by natural disasters can be determined through structural damage detection, which can aid rescue workers in prioritizing rescue and recovery operations. Local governments can use this information to effectively plan and distribute resources for recovery and reconstruction. In post-disaster environments, data gathering and analysis can be challenging, but a novel Deep Learning (DL) method for structural damage detection using aerial imagery has been proposed to address this problem. Overall, the field of structural damage detection is critical for ensuring the safety of people and the effective restoration of infrastructure following natural disasters. Advanced techniques such as vibration-based monitoring and Machine Learning (ML) algorithms have shown great promise in providing accurate and efficient solutions for detecting damage to buildings and other infrastructure. With further research and development, these techniques have the potential to revolutionize the way we approach structural damage detection and recovery efforts in the after-effects of natural disasters.

1.1.1 DEEP LEARNING

Buildings and infrastructure can sustain damage from quakes, storms, and floods, which can be detected via structural damage detection. DL has made substantial advancements in object detection, a crucial problem in computer vision and image processing. Structure-damage detection methods based on vibration have also been developed, and DL algorithms in particular have found increasing application in this field. ML methods that can be utilized to enhance model performance include active learning, grid search, and ensemble learning. Although acquiring and analyzing data in post-disaster settings is challenging, a novel DL approach using aerial imagery is offered for structural damage diagnosis. In general, DL has demonstrated outstanding performance in object detection and localization, and has the potential to improve structural damage detection in various industries.

1.1.2 MACHINE LEARNING

ML is an associated application of Artificial Intelligence (AI) that builds a mathematical model, using sample data which is known as training data to make predictions or decisions and provides systems the capability to learn automatically. However, the machine can learn without being programmed explicitly by any sources. ML focuses on programs that can access data and use them to learn for them. The process gets started by feeding good quality data and training systems by building machine learning models using data and different algorithms. The machine can learn through the pattern it forms while training, relate it with the training data, and thereby produce reliable, repeatable decisions and results. Every machine algorithm consists of three components - representation, evaluation, and optimization. Supervised and unsupervised algorithms are the two main categories in ML methods.

ML integrates several approaches such as combinatorial optimization, search, statistics, Reinforcement Learning (RL), probability theory, logic, and control theory. To create a good ML system, data preparation capabilities, basic or advanced algorithms, automation, ensemble modeling, and scalability are focused deeply. The computation algorithm plays a major role in the process of training.

1.1.3 OBJECT DETECTION

Object detection is a crucial task in computer vision and image processing, which involves detecting instances of semantic objects in digital images and videos. DL, particularly Convolutional Neural Networks (CNNs), has significantly advanced object detection by enabling accurate and efficient detection and localization of objects in various industries. Vibration-based techniques have been developed to evaluate the structural condition and detect structural problems, and ML algorithms, particularly DL algorithms, have become increasingly practical and widely used in vibration-based structural damage detection. Active learning , grid search, and ensemble learning are ML techniques that can improve the performance of DL models. Structural damage detection in post-disaster environments using aerial imagery is a challenging task, and a novel DL approach has been proposed to address these issues.

1.1.4 ACTIVE LEARNING

Active learning is a ML technique that involves selecting the most informative samples from a large dataset to be labeled by an expert. This process improves the performance of the model and helps reduce the amount of labeled data needed to train a DL model. By iteratively selecting the most uncertain or informative samples for labeling, the model can learn from the labeled data and improve its performance, while minimizing the overall cost of data labeling.

Active learning has shown promising results in various applications, including object detection, Natural Language Processing (NLP), and image classification, and can be particularly useful when data is limited or expensive to acquire.

1.1.5 ENSEMBLE LEARNING

Ensemble learning is a ML technique that involves combining multiple models to improve overall system performance. It can be used to explore the hyperparameter space for each model in the ensemble, and the best combination of hyperparameters can be selected for each model. This approach can lead to better generalization and improved accuracy compared to a single model. Grid search, a hyperparameter optimization technique, can be used to fine-tune the hyperparameters for each model in the ensemble. In addition, active learning can be used to select the most informative samples from a large dataset to be labeled by an expert, which can improve the performance of the ensemble model and reduce the amount of labeled data needed to train it.

1.1.6 UNMANNED AERIAL VEHICLES

Unmanned Aerial Vehicles (UAVs) have become increasingly popular in various industries, including disaster response and structural damage detection. UAVs equipped with cameras and sensors can capture aerial imagery of disaster-stricken areas and damaged structures, providing valuable information for rescue and recovery operations. With the advances in DL and computer vision techniques, UAVs can be used to automate the process of structural damage detection using aerial imagery, reducing the time and resources needed for manual inspections. Additionally, UAVs can be used to access hard-to-reach areas and provide real-time data to aid decision-making.

UAVs have the potential to revolutionize disaster response and structural damage detection, making them an essential tool for civil protection organizations and rescue teams.

1.2 OVERVIEW OF PROPOSED WORK

The proposed system aims to address the challenges of structural damage detection in post-disaster environments using UAVs and DL techniques. The system, called Feature-based Structural Damage Detection (FSDD), leverages the capabilities of UAVs to capture high-resolution images of affected areas and utilizes DL algorithms to automatically detect and assess the extent of structural damage.

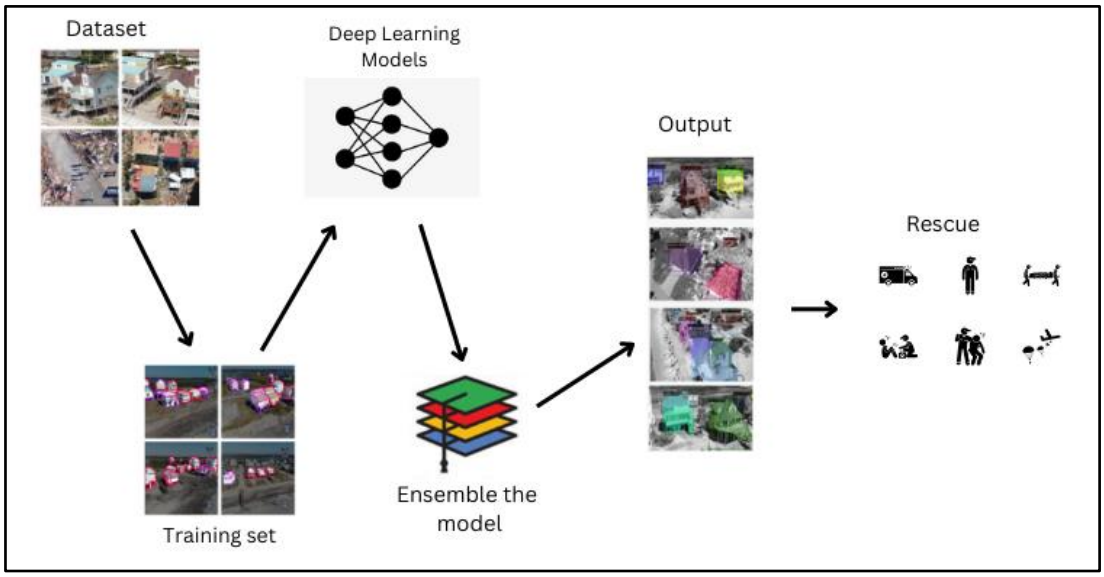

Fig. 1.1: Overall description of the proposed FSDD

Fig. 1.1 describes proposed FSDD technique. The FSDD technique is based on DL, which allows the system to learn and identify patterns in UAV imagery with a high degree of accuracy and reliability. By training the DL model on large datasets of labeled images, the system can improve its performance and generalizability across various disaster scenarios and types of buildings.

The use of UAVs for post-disaster damage assessment is advantageous due to their flexibility, mobility, and ability to capture detailed images of damaged structures. The FSDD technique takes advantage of these features to automate the damage detection process and reduce the risks associated with manual inspections, which can be dangerous and time-consuming.

The evaluation of Instance Segmentation in Building Damage Assessment (ISBDA) dataset demonstrates the effectiveness of the FSDD technique in detecting different types of structural damage, such as cracks, fractures, and displacements, with high precision and recall. Furthermore, the system can quantify the severity of the damage and provide a priority percentage for buildings that require immediate attention, aiding in the efficient allocation of resources during the response and recovery operation.

1.3 PROBLEM STATEMENT

To accurately assess structural damage in post-disaster environments. The proposed solution combines UAVs with DL to automate the damage detection process. By training a DL model on labeled UAV imagery, the system aims to provide timely and accurate information on the severity of damage, enabling effective response and recovery efforts.

1.4 OBJECTIVES

- To overcome the limitations of manual inspections by leveraging the flexibility, mobility, and high-resolution imaging capabilities of UAVs, combined with the accuracy and reliability of DL algorithms.

- Developing a system that automates the damage detection process, reducing the reliance on manual inspections and enabling faster assessment of structural damage in post-disaster environments.

- Utilizing DL techniques to train a model on labeled UAV imagery, enabling the system to accurately identify and classify various types of structural damage with high precision and recall.

- Providing timely and comprehensive information on the severity of damage, allowing for prioritization of response and recovery efforts to ensure the safety of individuals and efficient restoration of infrastructure.

CHAPTER 2

LITERATURE SURVEY

Quick responses to natural calamities greatly assist disaster relief efforts. However, current damage assessments primarily rely on human-monitored detection and reporting, which is a time-consuming, laborious, and costly process. This limits the effectiveness of disaster relief efforts and highlights the need for innovative solutions to improve the speed and accuracy of damage assessments.

The assessment of building damage following natural disasters can be carried out efficiently using MSNet [4], a multilevel segmentation network that analyzes aerial videos. This paper also introduces the ISBDA dataset, which contains annotated aerial videos from social media to serve as a benchmark for models that detect building damage. The Building Damage Detection Network-plus (BDDNet+) [5], a DL network, is proposed in this study for fast and accurate damage assessment of buildings. This network employs multiscale residual convolution blocks and self-attention layers to generate damage maps, and an XAI framework for interpretable and explainable results.

A novel approach [6] for detecting damaged buildings from post-earthquake remote sensing images is proposed, which utilizes multiple-feature analysis instead of relying on a single feature. This method evaluates both the building edge integrity and interior roof through a new feature parameter called edge significance and local gradient orientation entropy. Furthermore, this paper [7] examines images shared on social media platforms during natural disasters to assess the extent of damage caused by the disasters. By leveraging the rapid access to situation-sensitive data through social media networks, this study aims to address the time-consuming and challenging task of damage assessment during disasters, which typically takes weeks or months to complete.

In this article, an architecture [8] is proposed for assessing building damage in multitemporal remote sensing interpretation tasks. This approach aims to overcome the limitations of CNNs in modeling the nonlocal pixel relationship. The proposed method utilizes a siamese Transformer encoder to extract non-local and informative deep features from multitemporal image-pairs, followed by a multitemporal fusion module to integrate the information for downstream tasks. Furthermore, an end-to-end trainable model, RescueNet [9], is proposed to simultaneously segment buildings and assess damage levels of individual buildings in the aftermath of natural calamities using satellite and UAV imagery. This model aims to overcome the limitations of existing two-stage approaches that yield poor overall results and are not end-to-end trainable. For building segmentation, the paper proposes a binary cross entropy loss, followed by a selective categorical cross-entropy loss for categorizing damage, which enables modeling of the composite nature of the problem.

The use of satellite images [10] and building footprints before and after a battle can enable the calculation of damages to buildings. This paper proposes a multi-featured method that is highly robust to light changes and small deformations, which can effectively detect changes in buildings and describe building neighborhoods. By utilizing building footprints to assess changes in building height and form a neighborhood environment, the proposed method can accurately assess damages to buildings.

Furthermore, an unsupervised development of Improved Generative Adversarial Network (IGAN) model [11], which learns the crucial characteristics concealed within complex data by minimizing the divergence between the original response and the reconstructed response. Additionally, it minimizes the gap between the latent vectors obtained from two distinct encodings. During the testing phase, discrepancies in the latent features of the two codes are utilized to tentatively assess whether the structure is damaged.

The method proposed in [12] aims to identify earthquake-induced building damages using temporal Synthetic Aperture Radar (SAR) data acquired through different modes of observation. The authors propose a contextual change analysis technique based on novel textural features to overcome the limitation of using only SAR data acquired in the same observation mode. The objective is to improve the detectability of building damages in targeted areas while minimizing false alarms in agricultural regions. The proposed method is evaluated using grid-based accuracy analysis and aims to achieve a high detection rate with minimal false alarms, even on challenging datasets. In addition, this paper presents a road damage detection algorithm [13] that can accurately and efficiently detect damaged road areas without any parameters. The proposed technique transforms the disparity map by minimizing an energy function based on a road disparity projection model and stereo rig roll angle, facilitating the segmentation of the transformed disparity map using Otus's thresholding method. This approach enables the identification of damaged road areas.

In [14], a DL approach is introduced for detecting road damage, which seeks to enable quick assessments and repairs, decrease maintenance expenses, and extend the lifespan of asphalt. The authors use a varied dataset of crack distress types captured through mobile devices to train a set of scalable and efficient models that are specifically designed for pavement crack detection.

In [15], the Global Road Damage Detection Challenge (GRDDC) is discussed, which includes the top 12 solutions proposed by participating teams. Among these solutions, the most effective model leverages YOLO-based ensemble learning. In [16], a DL-based network is proposed for pavement crack detection, which employs the latest transformer module to capture long-range crack dependencies and improve detection performance. This model is trained using pavement datasets from India, the Czech Republic, and Japan, and has fewer network parameters. In [17], a numerical investigation is presented for the rapid detection of damaged bridges using a semi-supervised DL model and a

damage index-based Gaussian process. The study collects data from several damage scenarios within a two-axle moving vehicle load supported by a girder bridge. The proposed one-class CNN model incorporates a one-class classification algorithm with a one-dimensional CNN configuration.

In [18], a novel approach called mask scoring based on the Region-based Convolutional Neural Networks (R-CNN) network is proposed for instance segmentation. This technique aims to improve the predicted instance mask quality by addressing the issue of inconsistency between the mask quality and the instance classification score. The authors introduce a mask scoring strategy that aligns mask quality and score to achieve more precise mask predictions during COCO evaluation. Additionally, [19] introduces a new instance segmentation method called PolarMask, which is a single-shot approach that does not rely on anchor boxes and can be easily integrated into most detection methods. This method predicts the instance contour using a polar coordinate system and dense distance regression, with instance center classification.

A vehicle damage detection algorithm [20] using Transfer Learning (TL) and an enhanced Mask RCNN is proposed for swift resolution of traffic accident compensation issues. The proposed algorithm optimizes ResNet and feature extraction using Feature Pyramid Network (FPN), adjusts the Anchor's proportion and threshold in Region Proposal Network (RPN), maintains spatial information via bilinear interpolation in ROIAlign, and introduces different weights in the loss function. A fully-convolutional model for instance segmentation [21] with high accuracy and real-time processing speed is presented. The model splits instance segmentation into two parallel subtasks: generating a set of prototype masks and forecasting per-instance mask coefficients. Finally, instance masks are created by linearly merging prototypes with the mask coefficients..

CNNs based on mask regions [22] for object detection and instance segmentation in Very High-Resolution (VHR) remote sensing images are

proposed. The method generates bounding boxes and segmentation masks for all objects and their instances. To prevent precision loss, the proposed approach introduces precise pooling method, which computes the two-order integral based on the continuous feature map. The authors aim to improve object detection and promote instance segmentation for VHR remote sensing. They evaluate their proposed approach using the NWPU VHR-10 dataset and demonstrate its effectiveness.

CHAPTER 3

PROPOSED SYSTEM

3.1 OVERVIEW OF THE PROPOSED SYSTEM

The proposed system FSDD, involves ensembling three models that have been trained with different optimizers, and active learning is applied to each model in order to improve the accuracy and efficiency of the system. In addition, a feature extraction algorithm is used to enhance the damage predictions made by the system. This algorithm helps to identify relevant patterns and relationships in the data that might not be easily apparent otherwise, which lead to more accurate predictions. Overall, the proposed system is designed to improve the accuracy and robustness of damage predictions. By using a combination of ensembling, active learning, and feature extraction, the system is able to account for different strengths and weaknesses of individual models, select the most informative examples to train on, and identify important patterns in the data. This approach can be particularly useful in situations where data is limited or expensive to obtain. By maximizing the use of available data and improving the accuracy of damage predictions, the proposed system has the potential to help reduce costs and improve rescue operations.

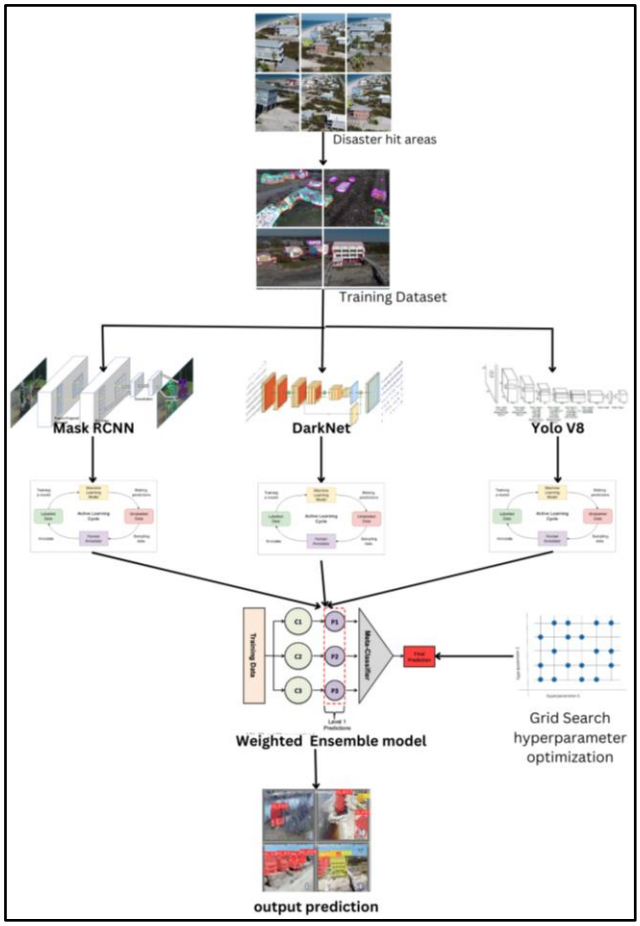

Fig. 3.1: Building Damage Detection using FSDD system

Fig. 3.1 shows the detailed flow of Building Damage Detection using FSDD system.

3.1.1 FEATURE EXTRACTION

Key features are extracted from images through analysis, including corners, edges, damages, and important points, and compared to extracted features across multiple images. One popular feature descriptor used in computer vision for feature detection and matching is the ORB descriptor. The ORB descriptor is a binary descriptor that is invariant to rotation and scale changes in the image, and is relatively fast to compute, making it suitable for real-time applications. The ORB descriptor is defined as the total sum of all binary tests conducted on a certain pixel position at different scales, with the tests comparing the intensity values of pairs of pixels located at a certain distance and orientation from the central pixel.

3.1.2 PREPROCESSING & ANNOTATION

For a comprehensive evaluation of building damages in diverse regions and environments, a dataset was assembled consisting of ten videos obtained from social media platforms. These videos were captured using aerial footage after several natural disasters such as Hurricane Harvey, Hurricane Michael, Hurricane Florence, and three tornadoes that occurred between 2017 and 2019. The affected areas were spread across different locations in the United States, including Florida, Missouri, Illinois, Texas, Alabama, and North Carolina, providing a broad range of areas to be analyzed.

The dataset [3] includes a total of 1,030 images extracted from ten videos, each annotated with instance-level building masks showing the damaged areas. The dataset comprises 2961 instances of damaged parts classified into three categories: Slight, Severe, and Debris. The preprocessing stage involved auto-orientation and resizing of the images to 640 x 640. Various data augmentation techniques like 25% saturation & exposure were applied to train the model to predict outputs in different scenarios. By utilizing this dataset, the model can make accurate predictions and perform well in diverse situations, improving its overall effectiveness in identifying building damages.

3.2 ARCHITECTURE OF THE FSDD SYSTEM

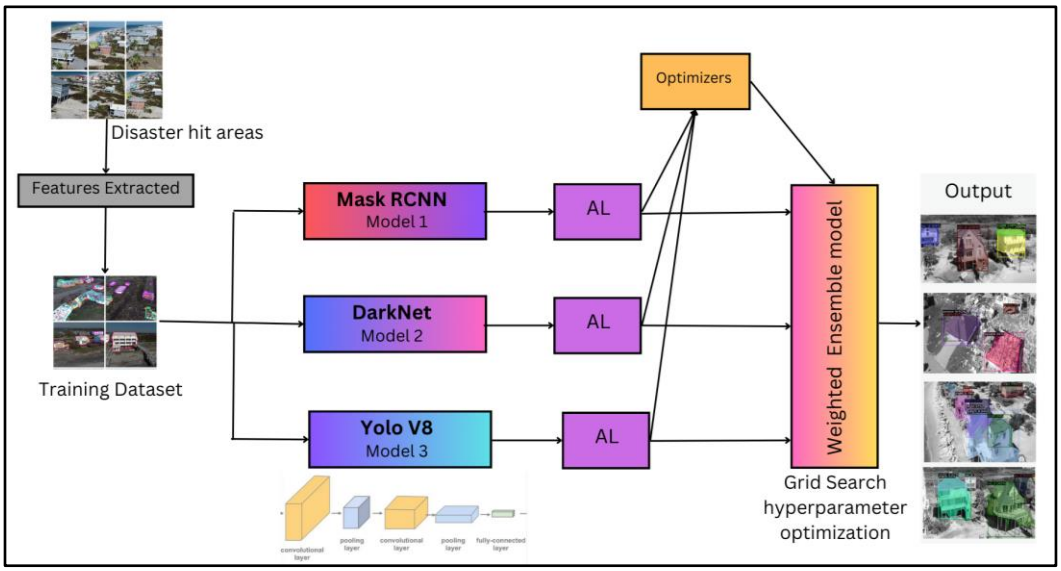

Fig. 3.2: High-level architecture of the FSDD system

Fig. 3.2 illustrates the working of the proposed FSDD system, wherein a YOLOv8, DETECTRON2 and YOLOv5 will be trained with the extracted features and active learning, optimizers are applied on each model and ensembled together which produced better results on ISBDA dataset than the existing works.

3.2.1 YOLOv8 CNN-MODEL

The architecture of YOLOv8 expands on the YOLO object detection models from earlier versions. The most recent version of the YOLO algorithm, known as YOLOv8, enhances earlier versions by adding new features like feature fusion, context aggregation and spatial attention modules. A fully convolutional neural network or backbone and head, is the method that YOLOv8 uses to process visual information. A modified version of the CSPDarknet53 architecture serves as the foundation of YOLOv8. 53 convolutional layers make up this architecture, which uses cross-stage partial connections to enhance information flow between the network's various levels. A number of convolutional layers and a string of fully connected layers make up the head of the YOLOv8 algorithm. These layers are in charge of determining the bounding boxes, class probabilities and object ratings for objects that are detected in an image.

3.2.2 DETECTRON2

Detectron2 is a comprehensive object detection and segmentation framework developed by Facebook AI Research (FAIR) that provides a flexible and modular platform for building and evaluating computer vision models. It is an open-source software library built on the PyTorch deep learning framework and offers a broad range of pre-trained models for object detection, instance segmentation, keypoint detection, and panoptic segmentation. These models are constructed using a combination of CNNs, ResNet architectures, and FPNs to achieve high accuracy and robustness. The RPN loss equation and FAST-RCNN loss equation is given as:

$$L_rpn = L_cls + \lambda * L_reg \quad \text{(Eq. 3.1)}$$

$$L_fast_rcnn = L_cls + \lambda * L_reg \quad \text{(Eq. 3.2)}$$

- L_cls is the classification loss, typically computed using binary cross entropy or sigmoid focal loss.
- L_reg is the bounding box regression loss, often calculated using smooth L1 loss.
- λ is a hyperparameter that controls the balance between the two loss components.

Moreover, Detectron2 includes a rich set of tools and utilities for data loading, data augmentation, model training, and model evaluation. These tools make it easy for users to customize and fine-tune models for specific datasets and use cases. Another key feature of Detectron2 is its modular architecture, which enables users to easily combine different components to create customized object detection pipelines. Additionally, Detectron2 supports distributed training, enabling users to scale their models across multiple GPUs or even multiple machines. Overall, these features make Detectron2 a powerful and versatile framework for developing and evaluating computer vision models.

3.2.3 YOLOv5:

Ultralytics developed YOLOv5, an object detection model that can accurately detect and locate objects in real-time from images or videos. The YOLOv5 backbone network is a modified version of the CSPDarknet53 architecture, which is also used in YOLOv4. This backbone comprises 53 convolutional layers with skip connections and is a deep convolutional neural network. YOLOv5 relies on anchor-based object detection, which enables it to predict bounding boxes and object classes directly from image pixels. The model leverages a fully CNN architecture, along with a combination of deep convolutional layers and residual connections, to detect objects accurately and efficiently in complex scenes. YOLOv5 is available in various versions, each with varying levels of speed and accuracy, giving researchers the flexibility to select the version that best suits their specific requirements.

In YOLOv5, each bounding box prediction is associated with an objectness score representing the presence of an object and class probabilities representing the likelihood of belonging to different object classes. These values are calculated using logistic regression and softmax functions, respectively.

$$Objectness\ Score = \sigma(t_obj) \tag{Eq. 3.3}$$

$$Class\ Probability = softmax(t_cls) \tag{Eq. 3.4}$$

$$bx = (\sigma(t_x) + cx) / grid_width \tag{Eq. 3.5}$$

$$by = (\sigma(t_y) + cy) / grid_height \tag{Eq. 3.6}$$

$$bw = anchor_width * exp(t_w) / grid_width \tag{Eq. 3.7}$$

$$bh = anchor_height * exp(t_h) / grid_height \tag{Eq. 3.8}$$

Here, σ denotes the sigmoid function, t_obj represents the objectness prediction, and t_cls represents the class predictions, bx and by represent the x and y coordinates of the center of the bounding box, bw and bh represent the width and height of the bounding box, sigmoid denotes the sigmoid function, exp denotes the exponential function, and cx, cy, anchor_width, and anchor_height are constants specific to each grid cell and anchor box.

3.3 ALGORITHM FOR THE PROPOSED FSDD SYSTEM:

Algorithm for feature extraction used in FSDD technique.

Feature Extraction and Ensemble models

Input: Collection of images S= {s1, s2, s3, ..., sn}
Output: feature descriptors, feature points, contours, vertices, and surface qualities

Initialization:

1. Load image from the input collection
2. Set up empty lists for feature points, feature descriptors, contours, vertices, and surface qualities

Function

```
detect_feature(I, x, y, s, angle):
        patch_size ← 20;
        patch ← I[int(y - patch_size/2) : int(y + patch_size/2), int(x -
        patch_size/2) : int(x + patch_size/2)];
        average_brightness ← patch.mean();
        if average_brightness < 150 then
                return True;
        else
        return False;

feature_points_list,
feature_descriptors_list←detectAndCompute(gray_scale_image, N_one);
features_list ← [];
for kp, desc in zip(feature_points_list, feature_descriptors_list) do
        x_coord, y_coord ← kp.pt;
        scale ← kp.size;
        orientation ← kp.angle;
        if detect_feature(gray_scale_image, x_coord, y_coord, scale,orientation)
        then
                features_list.append(desc);
        end
end
```

for each image in S:

1. Extract feature points, feature descriptors, contours, vertices, and surface qualities
2. Append the extracted features to their corresponding lists.

3.4 OPTIMIZATION AND ACTIVE LEARNING:

The process of active learning in training a model using a small quantity of labeled examples and selecting the most informative samples from a large dataset iteratively. This approach is employed to save money and effort on manual annotation. The process starts with training an initial model using a small quantity of labeled examples. The model is then used to predict the labels of the remaining unlabeled examples. A selection strategy is applied to choose the most informative examples based on their predicted labels, uncertainty, or diversity and add them to the training set. This iterative process continues until the model achieves a desired level of performance. Additionally, various optimizers such as Adagrad, Adam, and Stochastic Gradient Descent (SGD) are used during the training process to optimize the model's performance.

3.5 ENSEMBLE THE MODELS

Ensemble learning, which is a ML approach that combines the results of multiple models to improve the overall performance of a system. One such ensemble method is the voting ensemble, which aggregates the predictions of individual models using a majority vote or a weighted average. In this method, each model is trained on the same dataset using a different algorithm or set of hyperparameters, and generates its own set of outputs during prediction. These outputs are then combined using a weighted average or majority vote, with the most prevalent or heavily weighted output serving as the final prediction. The diversity and quality of each model can be improved by using grid search to find optimal hyperparameters for the ensemble as a whole or for individual models within the ensemble. The section can also discuss the benefits and limitations of ensemble learning and provide examples of its applications in various domains.

CHAPTER 4

IMPLEMENTATION

4.1 TOOLS AND LANGUAGES USED

4.1.1 PYTHON

Python is an interpreted, object-oriented, high-level, general purpose programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very suitable for rapid application development, as well as for use as a scripting or glue language to connect existing components together.

Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages, which encourages program modularity and code reuse. The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributed.

In this project, python has been used to model a YOLOv8, Detectron2 and Yolov5 CNN-based object detection model that will be able to detect structural damages in disaster hit areas.

4.1.2 OPENCV

OpenCV is a huge open-source library for computer vision, ML and image processing. It can process images and videos to identify objects, faces, and even human handwriting.

In this project, we use the OpenCV library to process the images present in the dataset, and to implement object recognition on them.

4.1.3 GOOGLE COLAB

Colab is a free Jupyter notebook environment that runs entirely in the cloud and supports collaborative development. It is a powerful platform for learning and quickly developing ML models in Python.

This project was programmed and stored in a Google Colab document.

4.1.4 CUDA

CUDA is a parallel computing platform that provides an API for developers, allowing them to build tools that can make use of Graphics Processing Units (GPUs) for general-purpose processing. ML processes large blocks of data, hence GPUs come in handy for ML tasks.

In this project, CUDA is used to enable a GPU for the program, which will improve the efficiency of training the CNN model.

4.1.5 MATPLOTLIB

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. It is a plotting library for the Python programming language and its numerical mathematics extension NumPy.

In this project, we use matplotlib to plot the average loss of the trained model with respect to the number of iterations through which the model has been trained.

4.1.6 ROBOFLOW

Roboflow is a cloud-based platform that provides a suite of tools for creating, training, and deploying custom object detection models for computer vision tasks. It offers features such as data management, model creation, training, deployment, and collaboration. With Roboflow, you can easily manage and prepare image datasets, create custom models using state-of-the-art algorithms, train them on large datasets, and deploy them to a variety of platforms.

4.2 EXPERIMENTAL SETUP

The training was done on the ISBDA dataset using Google Colab with 12 GB of RAM and 110 GB of disk space. The Google Colab environment was further equipped with a high performance NVIDIA Tesla K80 GPU. The ISBDA dataset was used to train the DL model. It was annotated using the Roboflow software and various preprocessing techniques are also applied. This dataset contains around 160 images taken from an UAV. 70% of these images were used to train the model, and the remaining 30% were used for testing the accuracy of the trained model. The YOLOv8 model was trained for 200 epochs, which was executed in 60 minutes. The images present in the dataset contained 5 classes, namely no_d, slight_h, slight, severe and severe_h. The dataset was balanced, as an equal number of images containing all the 5 classes were selected for the training and testing set.

4.3 EVALUATION METRICS

4.3.1 TRAIN & TEST SET

The training set is used to train a model, while the test set is used to estimate the model's performance on new, unseen data. The thing is to produce a model that can generalize well to new data, meaning that it can directly prognosticate issues on data that it has not seen ahead. The split between training and test data is generally determined aimlessly, with a typical split being 80 for training and 20 for testing.

4.3.2 mAP

Mean Average Precision (MAP) is a common metric used for object detection tasks. It measures the average precision over multiple classes and provides a summary of the performance of the model across all classes. It is calculated as the mean of the AP (Average Precision) values for each class.

4.3.3 PRECISION AND RECALL

Precision can be thought as a measure of exactness and recall (same as sensitivity) can be thought as a measure of completeness.

$$Precision=\frac{TP}{TP+FP}$$

(Eq. 4.2)

$$Recall=\frac{TP}{TP+FN}$$

(Eq. 4.3)

4.3.4 F1-SCORE

The F-Score, also called the F1-Score, is a measure of a model's accuracy on a dataset. It is used to evaluate binary classification systems, which classify examples into 'positive' or 'negative'.

$$F1\text{-}Score=\frac{2*Precision*Recall}{Precision+Recall}$$

(Eq. 4.4)

4.3.5 AVERAGE LOSS

The average loss of a model is the error obtained in the output produced by the model with respect to the ground truth. It is calculated for each iteration during the training of the model. A lower value for average loss is preferred.

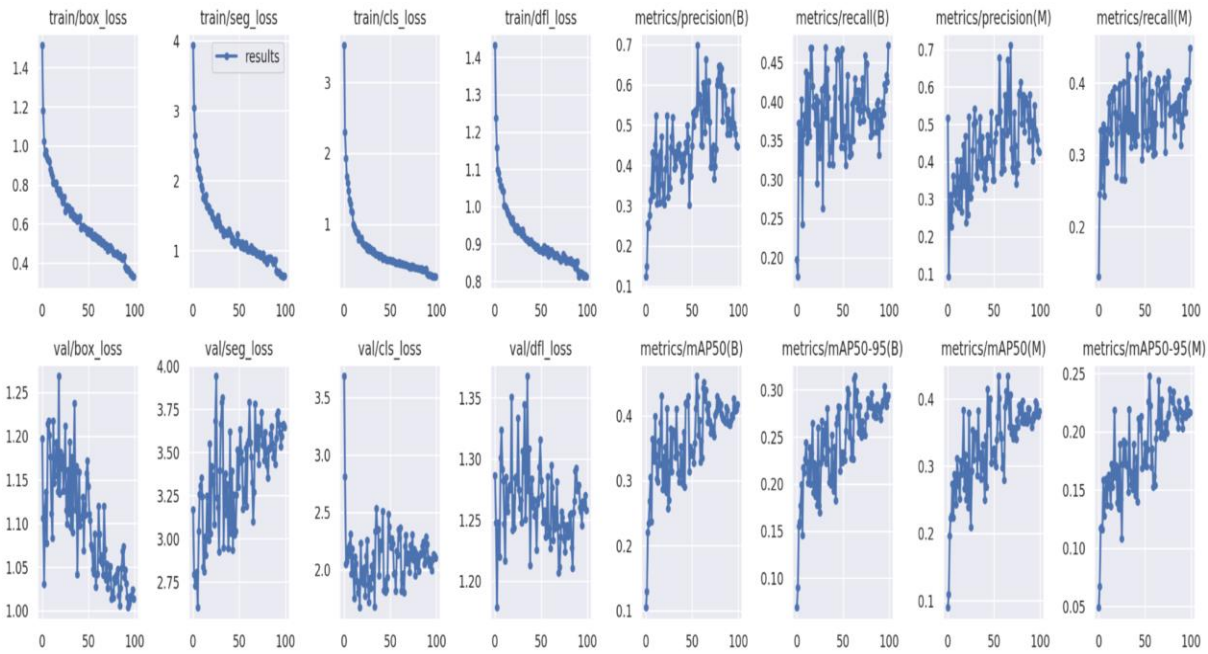

Fig. 4.1: Evaluation Metric.

The graph above illustrates the train loss, mAP@50-95, val loss, mAP@50, precision and recall of the YOLOv8 model as the number of iterations through which the model was trained increases as shown in Fig. 4.1.

4.4 COMPARISON OF THE PERFORMANCE OF VARIOUS DL MODELS & FSDD SYSTEM

In this component, the various models experimented are trained to analyze the various performance of models based on the dataset. Metrics like mAP, precision, recall, F1-score are recorded, tabulated, and visualized below for all the YOLOv8, Detectron2, YOLOv5 models and FSDD system.

Table 4.1 Performance comparison of various CNN & FSDD system

Model	**mAP**	**Precision**	**Recall**	**F1-Score**
YOLOv8	0.89	0.89	0.85	0.87
DETECTRON2	0.83	0.80	0.84	0.82
YOLOv5	0.84	0.82	0.84	0.83
FSDD	0.91	0.93	0.91	0.92

The metrics values are plotted for all the four models and shown below in the graph in Fig. 4.2. The precision, recall and mAP values were computed to be, 0.93, 0.91 and 0.91 respectively, for the FSDD model as shown in Table4.1. Precision is the ratio of the number of images correctly identified with the class present in the image, to the total number of predictions with that class. Sensitivity or True Positive Rate (TPR) or recall is a measure of the proportion of images predicted with a class, correctly identified with that class. The harmonic mean of precision and recall is the F1 score.

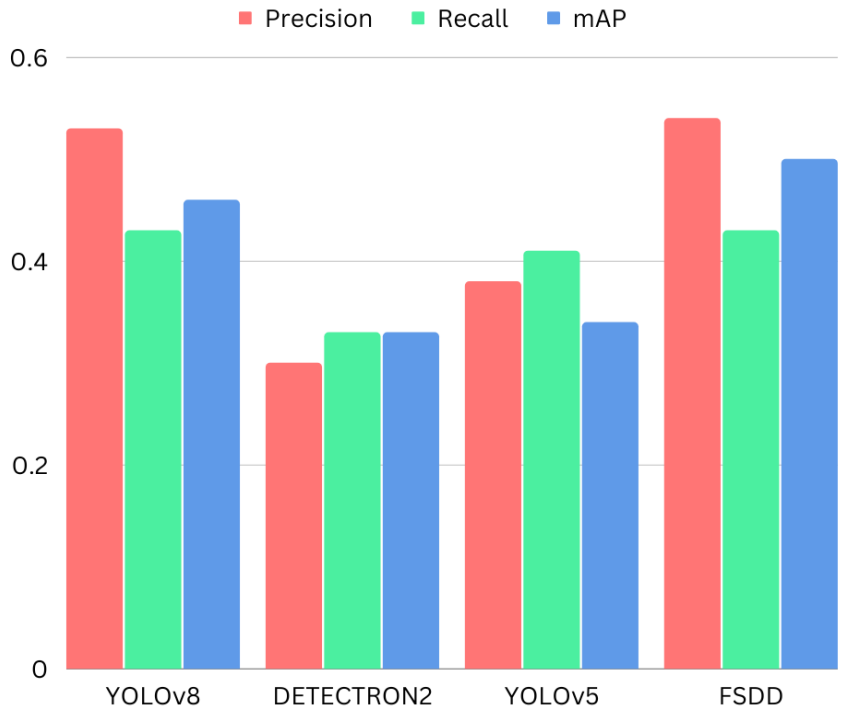

Fig .4.2 Performance Metrics Comparison

These metrics were calculated for other CNN models, namely Fast YOLOv8, DETECTRON2, and YOLOv5. From this plot, it is evident that the FSDD model has the highest mAP compared to other CNN models. The precision, recall and F1 score of FSDD were higher than that of the other existing works done using ISBDA dataset.

The order of the models based on their mAP is as follows:

FSDD> YOLOv8 > YOLOv5 > DETECTRON2

From the above comparison we can see FSDD provides better performance compared to other DL models.

CHAPTER 5

RESULTS AND ANALYSIS

Fig. 4.1 shows a graph of various evaluation metrics with the respect to the iterations through which the model was trained. The average loss function is the function defined on a data point, prediction, and label, which measures the penalty. It is the training loss or the error on the training dataset. The average loss of the model decreased with an increase in the number of iterations, thereby showing the model's ability to detect damages with good accuracy. From the plot, it was observed that the mAP of YOLOv8 was 0.89 and the ensemble of these using different optimizers and active learning yielded 0.91. The predicted images are shown in Fig .4.3 and Fig .4.4.

Fig. 4.3: Testing of the YOLOv8 Model.

Fig. 4.4: FSDD Model predicted result.

CHAPTER 6

FUTURE WORKS

There is still room for improvement, and future research can focus on enhancing the scalability and efficiency of these methods for large-scale damage assessment. Additionally, the integration of these techniques with other forms of data, such as LiDAR and SAR, can provide a more comprehensive understanding of damages in different environments. Overall, the findings of this research can serve as a stepping stone for the development of more advanced and effective approaches for damage assessment in the future.

CHAPTER 7

CONCLUSION

In conclusion, our work has presented various approaches for damage detection and assessment using DL-based methods in the fields of remote sensing and computer vision. The studies discussed in the work have addressed the limitations of traditional methods and have proposed novel techniques for detecting and segmenting damages caused by natural disasters and accidents. The empirical evaluations of these methods on different datasets have demonstrated their effectiveness in accurately identifying damages and reducing false alarms.

REFERENCES

[1] L. Jiao, F. Zhang, F. Liu, S. Yang, L. Li, Z. Feng, and R. Qu, "A survey of deep learning-based object detection," IEEE access, vol. 7, pp. 128837–128868, 2019.

[2] O. Avci, O. Abdeljaber, S. Kiranyaz, M. Hussein, M. Gabbouj, and D. J. Inman, "A review of vibration-based damage detection in civil structures: From traditional methods to machine learning and deep learning applications," Mechanical systems and signal processing, vol. 147, p. 107077, 2021.

[3] D. Agarwal, O. Covarrubias-Zambrano, S. Bossmann, and B. Natarajan, "Impacts of behavioral biases on active learning strategies," in 2022 International Conference on Artificial Intelligence in Information and Communication (ICAIIC), pp. 256–261, IEEE, 2022.

[4] X. Zhu, J. Liang, and A. Hauptmann, "Msnet: A multilevel instance segmentation network for natural disaster damage assessment in aerial videos," in Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp. 2023–2032, 2021.

[5] S. T. Seydi, M. Hasanlou, J. Chanussot, and P. Ghamisi, "Bdd-net+: A building damage detection framework based on modified coat-net," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2023.

[6] X. Ye, M. Liu, J. Wang, Q. Qin, H. Ren, J. Wang, and J. Hui, "Buildingbased damage detection from postquake image using multiple-feature analysis," IEEE Geoscience and Remote Sensing Letters, vol. 14, no. 4, pp. 499–503, 2017.

[7] H. Chen, E. Nemni, S. Vallecorsa, X. Li, C. Wu, and L. Bromley, "Dualtasks siamese transformer framework for building damage assessment," in IGARSS 2022-2022 IEEE International Geoscience and Remote Sensing Symposium, pp. 1600–1603, IEEE, 2022.

[8] R. Gupta and M. Shah, "Rescuenet: Joint building segmentation and damage assessment from satellite imagery," in 2020 25th International Conference on Pattern Recognition (ICPR), pp. 4405–4411, IEEE, 2021.

[9] F. Zhao, J. Bao, and D. Ming, “Battle damage assessment for building based on multi-feature,” in 2020 IEEE 5th Information Technology and Mechatronics Engineering Conference (ITOEC), pp. 57–60, IEEE, 2020.

[10] B. Kim and S. Cho, “Automated multiple concrete damage detection using instance segmentation deep learning model,” Applied Sciences, vol. 10, no. 22, p. 8008, 2020.

[11] Y. Luo, X. Guo, L.-k. Wang, J.-l. Zheng, J.-l. Liu, and F.-y. Liao, “Unsupervised structural damage detection based on an improved generative adversarial network and cloud model,” Journal of Low Frequency Noise, Vibration and Active Control, p. 14613484221150804, 2023.

[12] M. Kim, S.-E. Park, and S.-J. Lee, “Detection of damaged buildings using temporal sar data with different observation modes,” Remote Sensing, vol. 15, no. 2, p. 308, 2023.

[13] R. Fan and M. Liu, “Road damage detection based on unsupervised disparity map segmentation,” IEEE Transactions on Intelligent Transportation Systems, vol. 21, no. 11, pp. 4906–4911, 2019.

[14] S. Naddaf-Sh, M.-M. Naddaf-Sh, A. R. Kashani, and H. Zargarzadeh, “An efficient and scalable deep learning approach for road damage detection,” in 2020 IEEE International Conference on Big Data (Big Data), pp. 5602–5608, IEEE, 2020.

[15] D. Arya, H. Maeda, S. K. Ghosh, D. Toshniwal, H. Omata, T. Kashiyama, and Y. Sekimoto, “Global road damage detection: Stateof-the-art solutions,” in 2020 IEEE International Conference on Big Data (Big Data), pp. 5533–5539, IEEE, 2020.

[16] X. Xiang, Z. Wang, and Y. Qiao, “An improved yolov5 crack detection method combined with transformer,” IEEE Sensors Journal, vol. 22, no. 14, pp. 14328–14335, 2022.

[17] F. Yessoufou and J. Zhu, “One-class convolutional neural network (occnn) model for rapid bridge damage detection using bridge response data,” KSCE Journal of Civil Engineering, pp. 1–21, 2023.

[18] Z. Huang, L. Huang, Y. Gong, C. Huang, and X. Wang, “Mask scoring r-cnn,” in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 6409–6418, 2019.

[19] E. Xie, P. Sun, X. Song, W. Wang, X. Liu, D. Liang, C. Shen, and P. Luo, “Polarmask: Single shot instance segmentation with polar representation,” in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 12193–12202, 2020.

[20] Q. Zhang, X. Chang, and S. B. Bian, “Vehicle-damage-detection segmentation algorithm based on improved mask rcnn,” IEEE Access, vol. 8, pp. 6997–7004, 2020.

[21] D. Bolya, C. Zhou, F. Xiao, and Y. J. Lee, “Yolact: Real-time instance segmentation,” in Proceedings of the IEEE/CVF international conference on computer vision, pp. 9157–9166, 2019.

[22] H. Su, S. Wei, M. Yan, C. Wang, J. Shi, and X. Zhang, “Object detection and instance segmentation in remote sensing imagery based on precise mask r-cnn,” in IGARSS 2019-2019 IEEE International Geoscience and Remote Sensing Symposium, pp. 1454–1457, IEEE, 2019.

[23] D. Guanlin, “Research on semantic segmentation algorithm based on deep learning control tools,” in 2020 International Conference on Computer Communication and Network Security (CCNS), pp. 35–38, 2020.